

Briefing: Richard Goodman / apoth3osis — Meet-and-Greet

Date: May 2026 **Purpose:** Initial video call to explore formal verification collaboration for ultrametric/Bruhat-Tits quantum computing claims. **Status:** Richard initiated contact via QWAV website inquiry (April 2026). Email thread covers ~6 weeks of substantive technical exchange. **Primary Contact:** Richard Goodman, Apoth3osis Labs — rgoodman@apoth3osis.io **Website:** <https://www.apoth3osis.io> — “Paper → Proof → Code”

1. Meeting Purpose

Explore whether Richard Goodman can formally verify core QWAV claims in Lean 4, specifically:

1. **The strong triangle inequality produces passive error confinement** in Bruhat-Tits tree encodings
2. **A threshold theorem for ultrametric quantum error correction** — analogous to the surface code threshold theorem from the 1990s, but geometric rather than measurement-based
3. **Energy barrier scaling:** $E_{\text{barrier}} \propto q^d$ with tree depth

The strategic goal: Use formal verification (Lean 4 proofs) to bypass both peer review and physical lab experiments. A Lean proof that type-checks is unassailable — no reviewer, no gatekeeper, no credential check. This is the ultimate realization of the substance-first strategy.

The practical goal: A co-authored paper with machine-verifiable repository proving specific QWAV claims as theorems, using known physical parameters from published quantum computing experiments (Google, IBM, Rigetti, etc.) rather than requiring new lab data.

2. Who Is Richard Goodman

Item	Detail
Role	Founder, Apoth3osis Labs
Platform	HeytingLean — formal verification stack in Lean 4
Scale	4,649 Lean 4 files, 876,000+ lines, zero sorry in core
Pipeline	25-phase verified compilation pipeline reducing Lean 4's type checker to SKY combinators (S, K, Y)
SKY proof	18,000 lines, zero non-standard axioms
Architecture	Six-lens categorical transport: Laws of Form, Heyting, Tensor, Clifford, Graph, Geometric

Item	Detail
Collaborator	Vladimir Veselov (Russian mathematician, student of Kolmogorov) — ultrametric structures on $\mathbb{F}_2^n$ subspace lattices
Style	Formal verification absolutist: "Machine-checked or it does not count."
Track record	Formalized 6 of Veselov's papers. Found and corrected a Table 1 cross-validation discrepancy. Veselov accepted the correction — the formalization made his claims stronger.
Relevant paper	<i>Polygraphic Computation on a Kernel-Verified Substrate</i> (ResearchGate)
Relevant page	https://www.apoth3osis.io/paper-proof-code/padic-functional-decoupling

Three things to understand about Richard:

1. **He's not an academic gatekeeper.** He evaluates work by whether it formalizes, not by where it's published. The Veselov collaboration proves he works with independent researchers.
2. **He's been wrong before and changed his mind.** He initially thought Veselov's approach wouldn't work, shelved it for months, then realized it just needed a new language. He'll apply the same skepticism-then-rigor to QWAV claims.
3. **He's filing patents.** His other current collaborator insisted on patent applications. This means IP discussions are relevant and he understands the IP/formalization boundary.

3. The Strategic Opportunity

Why This Matters for QWAV

Current Situation	With Formal Verification
300+ self-published papers	Lean proofs that type-check — unassailable
Computational validation (Tier 0)	Formal verification (Lean 4) — stronger than simulation
"Trust my math"	"The proof compiles"
Gatekeepers can reject	No gatekeeper in a type-checker

Current Situation	With Formal Verification
Lab experiments needed for credibility	Formal proof + published experimental parameters suffice

The Leapfrog

The standard path for quantum error correction credibility is: > Theory → Peer Review → Experimental Validation → Acceptance

The Lean path is: > Theory → Formalization → Type-Check → Acceptance

This cuts out both peer review and lab experiments. It's the QWAV strategy pushed to its logical conclusion.

The Three Collaboration Shapes Richard Proposed

#	Shape	Commitment	Output	Richard's Preference
1	Formalistic transport	Lower	Co-authored paper + Lean repo proving specific QWAV claims	Preferred starting point
2	Ultrametric triangulation	Higher	Three-way research program: QWAV p-adic + Veselov $\mathbb{F}_2^n$ + HeytingLean topos	"Would not propose until I understand what your ultrametric work actually looks like"
3	Position paper	Lowest	Substrate computation: LoF-to-physics bridge drawing on Kauffman, QWAV framing, HeytingLean record	"Primarily positioning value"

Recommendation: Start with Shape #1 (formalization transport). It's the fastest path to a concrete output, builds the foundation for #2, and produces a citable, verifiable artifact.

4. What QWAV Brings to the Table

Technical Assets Ready to Share

Asset	Status	Relevance to Collaboration
Bruhat-Tits tree data structure	Implemented in Python (simulations/btree	Directly formalizable in Lean
Strong triangle inequality verification	0 violations in 15,000 trials	The core theorem to prove
Error confinement experiment (0A)	Tree LER=0 at depths 3+ for $p_{\text{err}} \leq 0.40$	Computational evidence backing the formal claim
Energy barrier scaling (0B)	$E_{\text{barrier}}(d) = 2^d$ confirmed	Formalizable as exponential lower bound
UQC Release document	57 numbered citations, full mathematical definitions	The authoritative technical reference
Technical Deep-Dive	Published thermal stability $\Gamma \approx 80$, specific predictions	Source material for formalization targets
300+ open-access papers	Zenodo, ResearchGate, SSRN	Background corpus for Richard to reference

The Core Theorem to Formalize

Threshold Theorem for Ultrametric QEC (informal): In a Bruhat-Tits tree encoding of depth d over $\mathbb{Q}_p$, with physical error rate p_{err} per leaf qubit, the logical error rate at the root satisfies $p_{\text{logical}} \leq C \cdot p_{\text{err}}^{k \cdot d}$ for some constants C, k determined by the tree structure and error model.

Corollary: There exists a threshold p_{thresh} such that for all $p_{\text{err}} < p_{\text{thresh}}, p_{\text{logical}} \rightarrow 0$ as $d \rightarrow \infty$. This threshold is geometric (determined by the strong triangle inequality), not measurement-based.

Published Experimental Parameters to Use

There’s no shortage of quantum computing papers with experimental data. The formal verification can be parameterized with real numbers from:

Source	Parameter	Value
Google (2023)	Superconducting qubit T_1, T_2	~20–100 μs
IBM (2024)	Gate fidelity (Heron)	~99.9%
Quantinuum (2024)	Trapped-ion 2-qubit gate fidelity	~99.8%
Various	Cryogenic cooling limits at mK	~50 μW at ~10 mK
Commercial	4 K cryocooler cooling power	~1 W

Key point: We don’t need to run experiments. We need to prove that IF physical error rates are below a threshold (using real published numbers), THEN the tree encoding provides fault tolerance. The conditional proof is what matters.

5. Collaboration Proposal (What to Offer)

Proposed First Deliverable

Title: *A Threshold Theorem for Ultrametric Quantum Error Correction: Formal Verification in Lean 4*

Content: 1. Formal definition of Bruhat-Tits tree encoding in Lean 4 2. Statement and proof of the strong triangle inequality for tree-encoded states 3. Derivation of error confinement: $p_{\text{logical}} \leq f(p_{\text{err}}, d)$ 4. Proof of exponential suppression with tree depth 5. Parameterization with published experimental data from superconducting qubit platforms 6. Machine-verified repository (zero sorry)

Output: Co-authored paper + Lean repository, publishable on arXiv/Zenodo.

Timeline: 2-3 months for first theorem. Incremental — each additional claim is a new module.

Roles

Role	Who	What
Theory	Rowan	Provide the mathematical claims, tree structure definitions, error model specifications, published experimental parameters
Formalization	Richard	Translate claims into Lean 4, prove them, build the verified repository
Joint	Both	Write the paper, agree on scope, handle IP boundaries

IP Boundaries

Richard is filing patents. QWAV has provisional patents. **Clarify before any code is written:**

- The mathematical claims (theorems) are Rowan's — they come from QWAV's published corpus
- The Lean 4 formalization code is jointly owned or licensed — discuss
- Any new theorems discovered during formalization are joint
- Patentable methods that emerge from the formalization need a written agreement
- **Suggestion:** Simple MOU stating: QWAV provides the claims, apoth3osis provides the formalization, co-authorship on the paper, IP from pre-existing work stays with originator, new IP is jointly owned.

6. Key Background from the Email Thread

The Philosophical Alignment

Topic	Rowan's Position	Richard's Position	Convergence?
Laws of Form as substrate	"I'm still finding my footing" — interested but exploring	LoF is the foundation of HeytingLean's nucleus infrastructure	Richard's infrastructure IS the LoF-to-physics bridge Rowan is looking for

Topic	Rowan's Position	Richard's Position	Convergence?
Golden ratio Φ	Topological resonance at Φ	Φ as nucleus fixed point (derived, not posited)	Strong convergence — same conclusion from different directions
Primes/integers as constructions	Yes — they're abstractions layered on deeper substrate	Concedes the ontological point but argues engineering relevance is separate	Productive disagreement — Richard accepts the ontology but argues crypto still works
Shor's algorithm	Theoretical artifact; QC infeasible at scale	HeytingLean positioned as QEC verification for IBM	Direct conflict — must be surfaced and scoped
Hardware vs. software	Hardware sets the bounds; software-based QEC is thermodynamic dead end	"It just required creating a new language for the software"	The core tension — but Richard has been wrong about this before (Veselov case)

Richard's Key Quotes

"Machine-checked or it does not count."

"If your program contains a formally grounded mechanism by which the substrate claim cashes out into falsifying predictions about quantum computing specifically, that is the paper I want to read."

"A real collaboration means your claims go through the machine. Some come through intact and are stronger for it. Some require axioms to be made precise before they can be stated formally. Some may be falsified. All three outcomes are wins from a research standpoint."

"My only other current formal collaborator initially approached me with nearly the same goal, convinced that his work required new hardware. But... it just required creating a new language for the software."

The Veselov Precedent

Vladimir Veselov is a Russian mathematician (student of Kolmogorov) working on: - Non-Expanding Cryptographic Protocol (NECP) - Ultrametric structures on $\mathbb{F}_2^n$ subspace lattices

Richard formalized 6 of Veselov's papers in Lean, found and corrected a Table 1 error, and Veselov joined the team. **This is the template for a QWAV collaboration.** Veselov's ultrametric work on $\mathbb{F}_2^n$ is directly adjacent to Rowan's ultrametric work on Bruhat-Tits trees — this is what Richard means by "ultrametric triangulation."

What Richard Already Knows About QWAV

- Rowan's axiomatic physics program (LoF + topological substrate + Φ resonance)
- The ultrametric quantum computing positioning (from qwav.net/ResearchGate)
- The Quantum Laws of Form paper
- The "hardware sets bounds" thesis

What Richard Does NOT Yet Know

- The Tier 0 simulation results (error confinement, barrier scaling) —**share these**
 - The specific, falsifiable predictions (5 from Technical Deep-Dive)
 - The full UQC architecture document (57 citations)
 - The specific theorem statements ready for formalization
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7. Meeting Agenda / Talking Points

Opening (5 min)

1. Thank Richard for the substantive email exchange — it's rare to find someone who engages at this level
2. Acknowledge the Veselov precedent — it's exactly the kind of adversarial rigor that strengthens claims
3. State the goal: explore whether formalization transport (Shape #1) can verify the core ultrametric QEC claims

What We've Built Since Last Contact (5-10 min)

1. **Tier 0 computational validation suite** — 9 Python files demonstrating error confinement and barrier scaling
2. **Key result:** Tree encoding gives zero logical error at depth 3+ for physical error rates up to 40%, while flat encoding fails (LER up to 15.2%)
3. **Key result:** Energy barrier scales as $E_{\text{barrier}}(d) = 2^d$, verified exhaustively for small depths
4. **Strong triangle inequality:** Zero violations in 15,000 random trials across primes 2, 3, 5
5. **Three applications submitted** (VSD, FRO, EWOR) — all written, all pending

The Core Theorem (10 min)

Present the threshold theorem concept:

In a Bruhat-Tits tree encoding with depth d , the logical error rate at the root is bounded by $C \cdot p_{\text{err}}^{k \cdot d}$, giving a geometric threshold for fault tolerance.

This is: - **Formally verifiable** — the mathematical objects (trees, probability bounds) are all definable in Lean - **Parameterizable with real data** — plug in published qubit error rates from Google/IBM/Rigetti - **Analogous to the surface code threshold theorem** — a known, respected class of results - **Publishable regardless of outcome** — if the bound doesn't hold, that's valuable too

The Ask (5-10 min)

1. **Would Richard be interested in formalizing this theorem as Shape #1 (formalization transport)?**
2. **What's the scope?** Start with the strong triangle inequality $\rightarrow$ error confinement proof. Expand to full threshold theorem if that closes.
3. **What's the timeline?** Richard's infrastructure is built — the Lean module would be `Generative/UniAxiomatic.lean`
4. **What are the IP terms?** Propose simple MOU: pre-existing IP stays with originator, co-authorship on paper, joint ownership of new theorems discovered during formalization
5. **What does Richard need from us?** Formal theorem statements, tree structure definitions, error model specifications, experimental parameter references

The Tension (5 min)

Address the elephant: Richard's HeytingLean is positioned as QEC verification for IBM. Rowan's thesis is that software-based QEC is a thermodynamic dead end.

- **Frame it honestly:** "We disagree on whether software-based QEC can scale. But we agree that formal verification of QEC claims is valuable regardless. If the ultrametric threshold theorem formalizes, it either strengthens your verification layer (if hardware QEC works) or provides an alternative path (if it doesn't). Either way, the proof is useful."
- **Scope the collaboration to where we agree:** The formalization transport (Shape #1) is about proving mathematical claims about tree structures. It doesn't require agreeing on IBM's trajectory.
- **Use Richard's own precedent:** He was wrong about Veselov. He might be wrong about hardware-vs-software.

Next Steps (5 min)

1. Rowan sends formal theorem statements within [X days]

2. Richard assesses formalizability and scopes the Lean module
 3. Agree on IP terms (MOU or simple email agreement)
 4. Schedule follow-up for detailed technical planning
 5. Target: first theorem formalized within 2-3 months
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8. Questions to Ask Richard

Technical

1. "What's the Lean 4 formalization of a tree data structure in your stack? Do you have a pre-existing Tree or Graph module we can build on?"
2. "How do you handle probabilistic claims in Lean? The threshold theorem involves bounds on error probabilities."
3. "What experimental parameters from the quantum computing literature have you already formalized? Qubit error rates, gate fidelities, etc.?"
4. "Vladimir's $\mathbb{F}_2^n$ ultrametric structures — how do they relate to Bruhat-Tits trees over $\mathbb{Q}_p$? Is there an existing formal connection?"

Collaboration

5. "The Veselov collaboration took about a year. What made it take that long, and what would make a QWAV formalization faster or slower?"
6. "You mentioned your other collaborator insisted on patents. How do you typically handle IP in these collaborations?"
7. "What's the co-authorship model? Does the Lean repository count as a publication in your view?"

Strategic

8. "You said you were wrong about Veselov — that you thought his approach wouldn't work for months. What changed your mind? What should I show you that would change your mind about hardware-vs-software?"
 9. "If the ultrametric threshold theorem formalizes, would you see it as strengthening the QEC verification layer (your commercial positioning) or as an alternative to it?"
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9. What to Have Ready

Before the Call

- ☐ Tier 0 simulation running and results handy (can screen-share the ASCII tables)
- ☐ UQC Release document open (for quick reference to specific citations)
- ☐ Technical Deep-Dive §5 (falsifiable predictions) ready to reference
- ☐ The specific theorem statement written out formally (not just informally)
- ☐ List of published experimental parameters with citations (Google T_1/T_2 , IBM gate fidelities, etc.)
- ☐ This briefing document open for reference

To Share After the Call

- ☐ Link to Tier 0 simulation repository (or send the Python files)
 - ☐ UQC Release document (full 57-citation version from Obsidian/releases)
 - ☐ Formal theorem statements (LaTeX)
 - ☐ Proposed MOU/IP terms
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10. Risks & Watch-Outs

Risk	Mitigation
Richard pushes software-over-hardware framing too hard	Acknowledge the disagreement, scope it out. "We don't need to agree on IBM's trajectory to prove a theorem about tree structures."
Formalization reveals a flaw in the core claim	This is actually a win — better to find it in Lean than in a lab or a funding rejection. Frame it as "the formalization is working exactly as intended."
Richard claims the theorem is already known or trivial	He might — the strong triangle inequality is mathematically straightforward. The novelty is the application to QEC, not the inequality itself. Be ready to pivot: "Then formalization should be fast, and we can move to the harder claims."
IP conflict — Richard patents the formalized theorem	Pre-agree that mathematical theorems are not patentable (they're discoveries, not inventions). Implementation methods might be. Get this in writing.
Richard wants co-authorship on everything	Reasonable for the formalized theorems (he did the work). Not reasonable for the underlying QWAV theory (that's pre-existing). Scope the co-authorship to the specific formalization paper, not the QWAV corpus.

Risk	Mitigation
The collaboration absorbs too much time	Richard does the Lean coding (it's his infrastructure). Rowan provides theorem statements and reviews. This is the right division of labor.

11. IP & Terms — Suggested Starting Position

Proposed MOU Principles:

1. Pre-existing IP: Each party retains all rights to their pre-existing intellectual property. QWAV's mathematical claims and published corpus remain QWAV's. HeytingLean infrastructure remains apoth3osis's.
2. Joint Work: Any new theorems, proofs, or methods discovered during the formalization process are jointly owned. Commercial use requires mutual agreement.
3. Publication: Co-authorship on the formalization paper. Rowan provides the mathematical claims and experimental parameter analysis. Richard provides the Lean 4 formalization and verification.
4. Patentable Methods: If the formalization produces patentable methods (implementation techniques, not mathematical theorems), the parties negotiate commercialization terms in good faith. Mathematical theorems per se are not patentable subject matter.
5. Open Access: The Lean repository and paper are published open-access (arXiv, Zenodo). The proofs are publicly verifiable.
6. Term: Initial collaboration for one formalization target (threshold theorem). Extendable by mutual agreement.

12. Post-Meeting Follow-Up

When	Action
Same day	Send thank-you email with summary of agreed next steps
Same day	Update SPRINT.md with new collaboration status
Within 3 days	Send formal theorem statements + experimental parameter references
Within 1 week	Agree on IP terms (email confirmation or MOU)
Within 2 weeks	Richard provides initial Lean formalization assessment
Within 1 month	First theorem statement checked in Lean (even if partial)

Appendix A: Key Quotes from Richard

"A real collaboration means your claims go through the machine. Some come through intact and are stronger for it. Some require axioms to be made precise before they can be stated formally. Some may be falsified. All three outcomes are wins."

"I have reviewed your work and understand your objective. My only other current formal collaborator initially approached me with nearly the same goal, convinced that his work required new hardware. It took a lot of convincing... But, I was wrong; it just required creating a new language for the software."

"If your program contains a formally grounded mechanism by which the substrate claim cashes out into falsifying predictions about quantum computing specifically, that is the paper I want to read."

"Machine-checked or it does not count."

"The ultrametric triangulation... a three-way structural connection between your p-adic methods, Vladimir's $\mathbb{F}_2^n$ NECP constructions, and HeytingLean's topos infrastructure."

Appendix B: Technical Assets to Reference

Asset	Location	What It Contains
UQC Release	Obsidian/releases/2026/05/ Quantum Computation.md (external research release)	57 citations, full mathematical definitions, 6 testable predictions
Technical Deep-Dive	strategy/Technical Deep-Dive - Ultrametric Quantum Computing and AI.md	$\Gamma \approx 80$, $E \propto q^d$, Bi-2212 substrate
Tier 0 Simulation	simulations/	Error confinement (0A), barrier scaling (0B)
Honest Investment Assessment	strategy/Honest Investment Assessment - The 100K Question.md	Self-assessment, gaps, computational path
External Sources	strategy/External Sources and Citation Map.md	Source strength matrix, citation map
Richard's boundary draft	https://www.apoth3osis.io/	Visual intuition for his approach
Richard's polygraphic paper	ResearchGate: 404363291	His most recent technical publication

Briefing prepared 2026-05-11. Based on email thread (April–May 2026) and full QWAV library review. Update after the call with action items and decisions.